

Tannehill Precision

There is no revenue line in this project, which is why nobody at Tannehill Precision could agree on what it was worth.

Tannehill was a contract machining shop in Rockford, Illinois, running aerospace and heavy-equipment components on 41 machines across two buildings. The proposal was to replace three manually loaded milling cells with a single automated cell: robotic loading, in-process gauging, and lights-out running on the second and third shifts. Installed cost, \$4.2 million. Expected life eight years, salvage value \$400,000.

Marisol Quintero, the plant manager, had built the case on labor. The three manual cells were staffed by eleven operators across the shifts, at \$68,000 fully loaded each. The automated cell needed two, and Tannehill's practice was not to lay people off — the other nine would be redeployed into the grinding department, which had two open requisitions it had not been able to fill and had for eight months been paying overtime for more hours than seven people would work in a year.

Ahmed Baig in finance had pushed back on that in the review. "If we redeploy them, we still pay them. Where is the saving?"

Quintero's answer was that the grinding overtime stopped and the two open requisitions closed, which was nine people's worth of work Tannehill would otherwise have had to buy. Baig's answer was that this was a statement about the grinding department, not about the milling project, and that if grinding needed nine people it should ask for nine people.

Tannehill had frozen headcount at the prior year's level eighteen months earlier. The freeze covered requisitions; it did not cover overtime, which came out of each department's own operating budget. The two grinding requisitions predated the freeze and had survived it. Nothing in the freeze prevented moving people between departments.

There were four other items in dispute.

Scrap. The manual cells ran 3.4 percent scrap against \$22 million of annual material throughput. The automated cell's in-process gauging was quoted at 1.6 percent. Maintenance would rise by \$85,000 a year for the robotics service contract.

The old machines. Three of them, net book value \$310,000 combined, with a firm offer from a used-equipment dealer at \$540,000.

The floor space. The automated cell would go into the northeast corner of building two, which Tannehill currently rented to a customer as finished-goods storage for \$60,000 a year. The storage agreement had three years left and the customer had said informally that they would not renew — they were building their own facility. Tannehill’s cost accounting charged the milling department \$130,000 a year of plant overhead based on square footage occupied, and the automated cell occupied slightly less square footage than the three machines it replaced.

The do-nothing case. Tannehill’s maintenance engineer had flagged that two of the three manual cells would need spindle and way-system rebuilds in year four at roughly \$900,000 combined if they stayed in service. The controller had been asked how that would be treated and had answered without much hesitation: a rebuild of that size restores the machines to a materially better condition and extends their service life, so it would be capitalized rather than expensed, and depreciated on the same seven-year MACRS schedule beginning in year four. Two of those years fall outside the project horizon; whatever basis remained would be written off when the machines were retired at the end of year eight.

Tannehill’s tax rate was 25 percent and its cost of capital 11 percent. The automated cell is seven-year MACRS property. The \$400,000 is an estimate of what the cell would fetch at the end of year eight; it does not enter the depreciation schedule.

EXHIBIT 1 • ITEMS IN DISPUTE

ITEM	AMOUNT	TIMING
Automated cell, installed	\$4,200,000	Year 0
Salvage value	\$400,000	Year 8
Operators freed from the three manual cells	11 at \$68,000	Annual
Retained to run the automated cell	2	—
Redeployed to grinding	9	—
Grinding: unfilled requisitions	2	—
Grinding: overtime being paid	> 7 people-years	Annual
Material throughput	\$22,000,000	Annual
Scrap rate, manual cells	3.4%	—
Scrap rate, automated cell	1.6%	—
Incremental maintenance	\$85,000	Annual
Old machines: net book value	\$310,000	—
Old machines: dealer offer	\$540,000	Year 0
Tannehill Precision • MGMT 648 mini case		2
Storage rental income forgone	\$60,000	Years 1–3
Plant overhead charged to milling	\$130,000	Annual
Reduced labor from automated cell	\$220,000	Years 1–3

EXHIBIT 2 · SEVEN-YEAR MACRS

YEAR OF SCHEDULE	1	2	3	4	5	6	7	8
Percentage	14.29	24.49	17.49	12.49	8.93	8.92	8.93	4.46

The schedule runs one year longer than the class life because of the half-year convention. It ignores salvage value: the asset is written down to zero regardless of what it is expected to fetch.

ASSIGNMENT

- 1 Build the incremental after-tax cash flow for years 0 through 8. Handle the sale of the old machines correctly, including tax, and mind what MACRS has done to the book value of the automated cell by the time it is sold.
- 2 Quintero and Baig disagree about the nine redeployed operators. State the condition under which each is right, and say which one you believe given what the case tells you about grinding.
- 3 The floor space is already owned and already paid for. What, if anything, does the project owe for it — and for how long?
- 4 What happens to the \$130,000 plant overhead charge, and what happens to it in the department's reported cost per hour after the project? Are those the same question?
- 5 Compute NPV and IRR. Then say which single assumption you would most want to verify before spending \$4.2 million, and how you would verify it.

Before class

- READ** Eight minutes is enough. Come with a number, not a summary.
- BRING** Your cash flow schedule, your NPV, and the one assumption you are least sure of.
- EXPECT** Twenty minutes of argument. More than one answer is defensible; an undefended one is not.